

Are Small LLMs Ready for Coding Agents?

A tight harness connects controllable models to accepted full workflows.

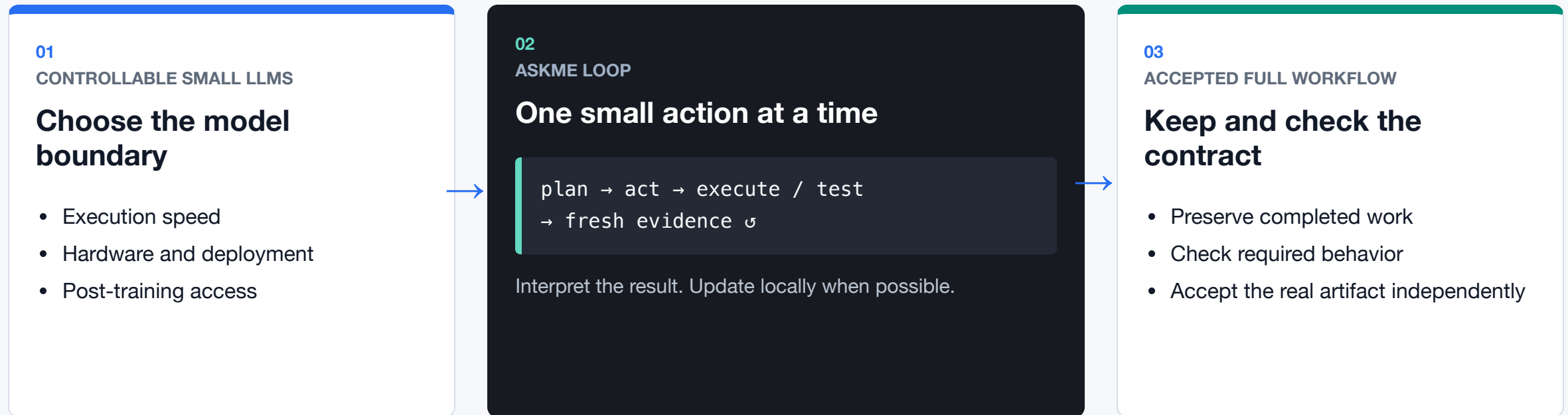
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A tight loop connects model control to full workflows

The model gets one bounded move; the harness keeps the end-to-end contract.



The bridge: small action surface + fresh execution feedback + independent workflow acceptance.

The code ran. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

Source and header written

The requested program was correct.

`/tmp/test` compiled

A different output path was chosen.

`/tmp/test` ran successfully

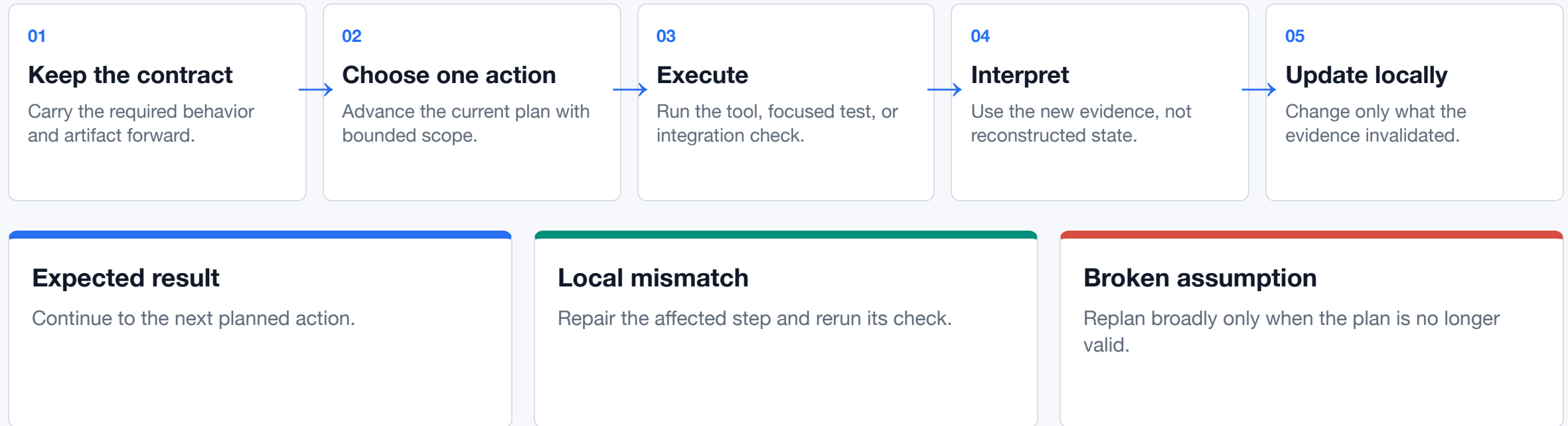
It printed `REPLAN_OK`.

Agent reported completion

The independent acceptance test found no `./main`.

Every recorded action succeeded. The deliverable contract still drifted.

Reason over fresh evidence. Preserve progress.



Trajectory goal: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn—not longer monologues.

Four models, eight descriptive receipts

Two variants per family; architecture and active parameters changed.

8 / 8 agent complete

7 / 8 artifact accepted

Hosted · simple checks · n=1/cell · descriptive only

MODEL	SHAPE	BUILD OBSERVATION	REPAIR OBSERVATION
Gemma 4 26B A4B	♦ MoE 25.2B total · 3.8B active	ACCEPTED 603.6s	ACCEPTED 20.0s
Gemma 4 31B	• Dense 30.7B parameters	ACCEPTED 66.5s	ACCEPTED 22.4s
Qwen3.6-27B	• Dense 27B parameters	ACCEPTED 47.9s	ACCEPTED 23.0s
Qwen3.6-35B-A3B	♦ MoE 3B active	NOT ACCEPTED wrong artifact path · 17.7s	ACCEPTED 11.8s

Boundary: architecture, active parameters, run order, and trajectories changed; the 31B follow-up was post-hoc. No size, family, reasoning, or reliability inference.

Freeze the smallest clean test

01

4 frozen workflows

Valid multi-file code, semantic failures, visible feedback, held-out scoring.

02

2 frozen policies

Explicit-reasoning off versus the current composite gated policy.

03

3 randomized repeats

$4 \times 2 \times 3 = 24$ scheduled runs per model.

04

2 primary outcomes

Held-out acceptance and false completion across all valid runs.

Secondary: repeated actions · work redone · local/full replans · latency · tokens

ANSWER TO THE TITLE

Promising—with a tight harness.

Model choice

Simple contract

Execution / test

Local reasoning

Accepted workflow

**Make the interface easier to use.
Keep success tied to real behavior.**

The smoke validates this interface; readiness needs repeated held-out workflows.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data